Villani Code Terminal-Bench Report

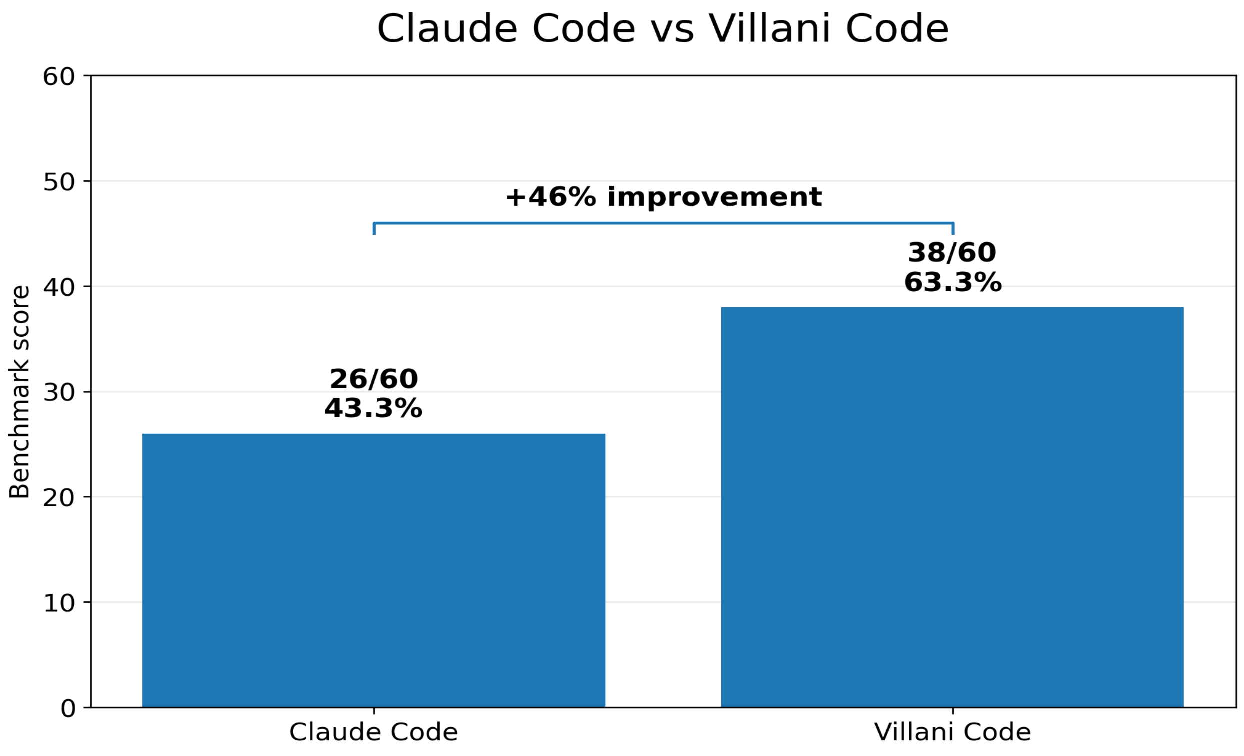
12-task Claude Code comparison and Qwen3.5 9B lower-bound suite score

38/60	26/60	+46%	99/445
Villani Code score	Claude Code score	relative performance improvement	current lower-bound score

Executive summary

Villani Code materially outperformed Claude Code when both were run with the same Qwen3.5 9B model on the same 12 overlapping Terminal-Bench tasks. Villani Code scored 38/60 versus Claude Code at 26/60, a 46% relative performance improvement and a 20 percentage-point absolute gap.

The comparison is especially useful because it strips away the foundation-model advantage. The model is held constant. The difference is the coding-agent runtime: execution loop, tool handling, state management, failure recovery, local model integration, benchmark reliability, and the new task-scoped memory system in the upgraded Villani Code branch.



Qwen3.5 9B - 12 overlapping Terminal-Bench tasks - 5 runs per task - Villani Code: 38/60 vs Claude Code: 26/60

1. Benchmark comparison

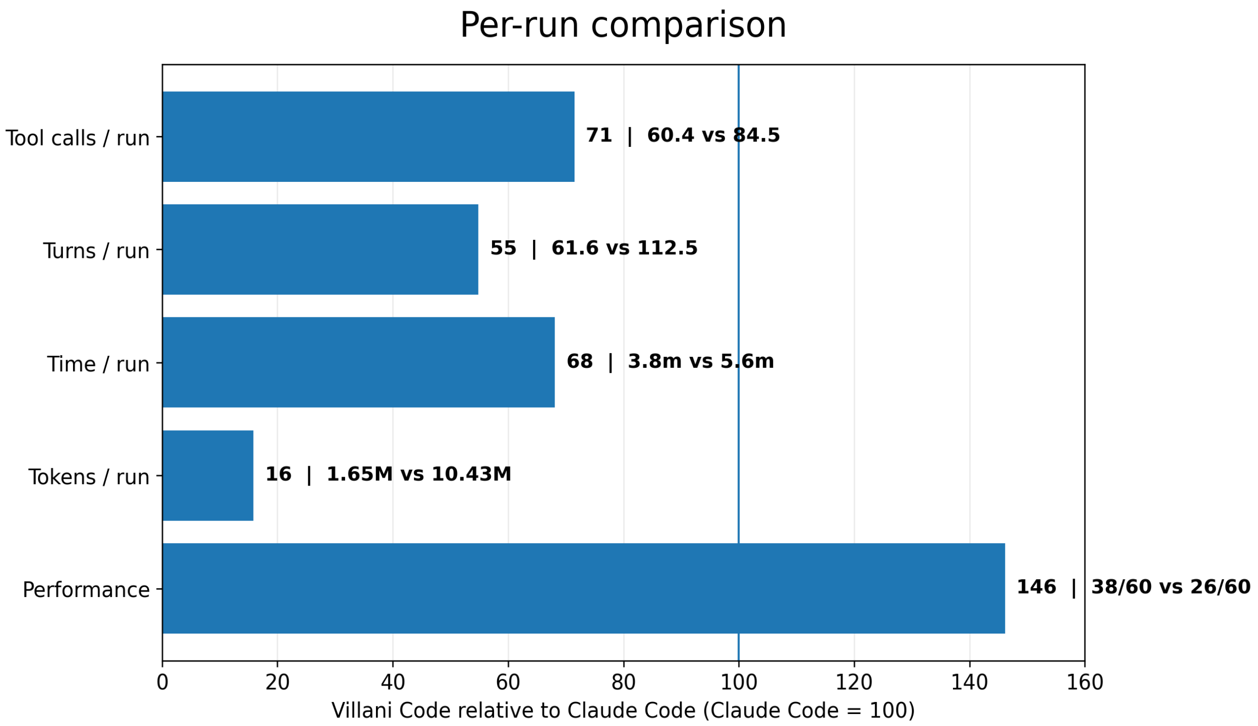
The 12-task overlap contains tasks that both Claude Code and Villani Code ran with Qwen3.5 9B. Each task was run 5 times, for 60 runs per agent. The result was 38 successful Villani runs against 26 Claude Code runs.

Metric	Villani Code	Claude Code
Score	38/60	26/60
Success rate	63.3%	43.3%
Relative performance	+46.2%	baseline
Task wins / ties / losses	6 / 6 / 0	0 / 6 / 6

Task	Villani	Claude	Delta
build-pmars	4/5	1/5	+3
fix-git	5/5	2/5	+3
git-multibranch	3/5	1/5	+2
kv-store-grpc	5/5	5/5	+0
mcmc-sampling-stan	3/5	3/5	+0
nginx-request-logging	5/5	4/5	+1
openssl-selfsigned-cert	4/5	4/5	+0
pypi-server	2/5	2/5	+0
query-optimize	2/5	0/5	+2
fix-code-vulnerability	1/5	1/5	+0
largest-eigenval	2/5	1/5	+1
sanitize-git-repo	2/5	2/5	+0

2. Per-run efficiency

The upgraded Villani Code runtime did not win by spending more. It achieved the higher benchmark score while using far fewer tokens, fewer turns, fewer tool calls, and less wall-clock agent time per run.

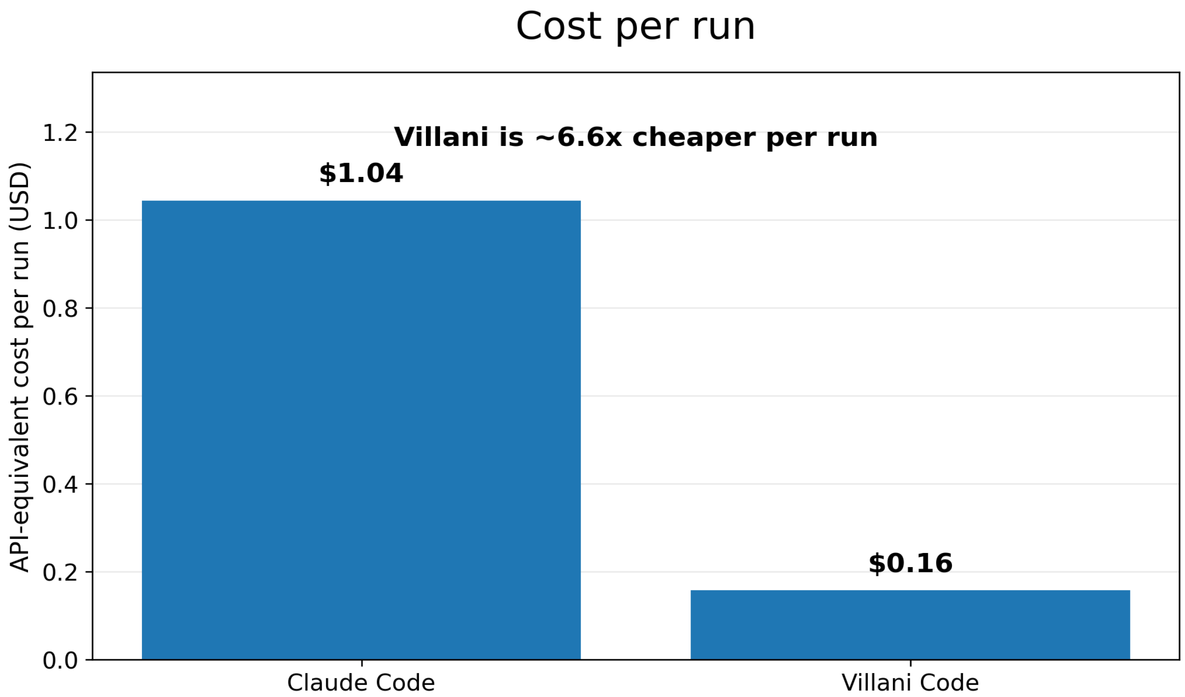


Qwen3.5 9B - 12 overlapping Terminal-Bench tasks - 5 runs per task - per-run values are means across 60 runs per agent

Metric	Villani Code	Claude Code
Mean tokens / run	1.65M	10.43M
Mean time / run	3.8 min	5.6 min
Mean turns / run	61.6	112.5
Mean tool calls / run	60.4	84.5

3. API-equivalent cost

Using Qwen3.5 9B list pricing of $0.10 per 1M input tokens and $0.15 per 1M output tokens, Villani Code is dramatically cheaper on the 12-task overlap. Cached tokens in Claude Code traces are counted as billable input tokens for this estimate.

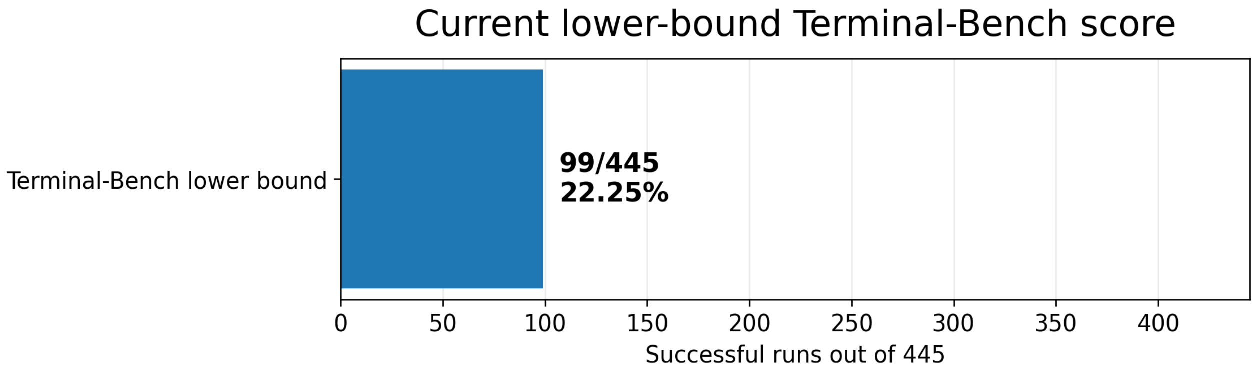


API-equivalent estimate using Qwen3.5 9B list pricing: 0.10/*Minputtokens*, 0.15/M output tokens. Cached tokens counted as input.

Metric	Villani Code	Claude Code
Total API-equivalent cost	$9.44	$62.62
Average cost / run	$0.16	$1.04
Cost / successful run	$0.25	$2.41
Relative cost / run	baseline	6.6x Villani

4. Current lower-bound Terminal-Bench score

Across the full 89-task Terminal-Bench suite, with 5 runs per task, the current lower-bound score for Villani Code with Qwen3.5 9B is 99/445. This is a lower bound because missing or still-unrun positive outcomes are not assumed. A strict clean-only variant, excluding the timed-out successful verifier result, is 98/445.



Qwen3.5 9B - lower bound across 89 tasks x 5 runs. Strict clean-only variant: 98/445.

Score definition	Score	Rate
Verifier-counted lower bound	99/445	22.25%
Strict clean-only lower bound	98/445	22.02%

Interpretation

The clean product story is simple: same small model, different coding agent, materially different results. Villani Code delivered a 46% performance improvement over Claude Code on the 12-task overlap while also reducing API-equivalent cost per run by roughly 85%.

This supports the Villani Code thesis: small local models do not just need better weights. They need a better runtime around them. When the model is held constant, agent runtime design becomes visible.

Method notes

All 12 comparison tasks were scored as 5-run task-level attempts using Terminal-Bench verifier outcomes. API-equivalent cost uses token telemetry from run traces. OpenRouter listed Qwen3.5 9B at $0.10/M input and $0.15/M output at the time this report was produced. The benchmark comparison does not isolate which individual Villani upgrade caused the improvement; it evaluates the upgraded runtime as a whole.